

The framework and memristive circuit design for attention-regulated working memory

Jihong Zhang^{a, b, c} , Xiaoping Wang^{a, b, c, *} , Zhanfei Chen^{a, b, c}, Chao Yang^d

^a School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, 430074, China

^b The Key Laboratory of Image Processing and Intelligent Control of Education Ministry of China, Wuhan, 430074, China

^c Hubei Key Laboratory of Brain-Inspired Intelligent Systems, Wuhan, 430074, China

^d School of Computer Science and Technology, Guangxi University of Science and Technology, Liuzhou, 545006, China

HIGHLIGHTS

- Novel working memory model integrates multisensory inputs and memory retrieval.
- Bio-inspired dual attention pathways drive goal-directed decisions.
- Memristive circuit achieves rich cognitive functions with online in-situ learning.
- Enables robots to handle stimulus loss and dynamic contexts through cognitive control.

ARTICLE INFO

Communicated by K. Liu

Keywords:
Memristor
Working memory
Attention
Bionic circuit
Cognitive control

ABSTRACT

Adaptive behavior and decision-making depend on constantly selecting relevant information from the external environment and internal states. Inspired by the working memory structure and the top-down and bottom-up attention mechanisms in cognitive neuroscience, this work proposes an attention-regulated working memory model. This model provides a brain-inspired approach to integrate perception, long-term memory, and action. It processes current external multisensory stimuli and retrieves stored knowledge from internal reinforcement simultaneously, leading to adaptive and rapid executive actions. On this basis, a memristive circuit is designed to realize rich cognitive functions in an online in-situ learning and in-memory computing manner. The designed circuit consists of four main components: (1) the phonological loop and visuospatial sketchpad consider different audio-visual input patterns and varying stimulus salience, realizing the filtration, synchronization, and encoding of multimodal signals; (2) the attention control module captures and maintains attention driven by multisensory stimulation; (3) the episodic buffer achieves reward reinforcement, forming or resetting the top-down attentional bias signal; (4) the central executive control module regulates the relationships between the two attentional pathways, thus transforming the random exploration process into a learnable final action. Finally, simulation results in LTSPICE demonstrate that our circuit can be adaptively applied to the cognitive control and execution system of robots within complicated circumstances.

1. Introduction

To navigate the complex world, we need to constantly choose information from our surroundings and memories. This is known as cognitive control or executive function, a set of psychological processes that utilize perceptions, knowledge, and goals to decide which actions to take from many options [1]. Such actions must be rapid, flexible, and adaptable,

allowing us to focus on task-relevant information and accommodate unforeseen changes. Memory plays a vital part in developing human learning, perception, decision-making, and other behaviors [2]. In human memory hierarchy, working memory (WM) can load the knowledge or experience from long-term memory (LTM) and process audio-visual signals from sensory memory (SM), which bridges the LTM and SM [3,4]. WM focuses on temporarily holding relevant information, whether it

* Corresponding author at: School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, 430074, China.

Email addresses: m202373626@hust.edu.cn (J. Zhang), wangxiaoping@hust.edu.cn (X. Wang), chenzhanfei@hust.edu.cn (Z. Chen), yangchao@gxust.edu.cn (C. Yang).

is from a recent experience or currently in the environment. It provides an interface that links perception (especially auditory and visual), LTM, and action, thus enabling goal-oriented behavior and executive decision-making [5].

Furthermore, we voluntarily attend to things that are relevant to our goals, but our attention may also be captured involuntarily by salient events in the external world. Attention influences how humans process sensory information, store it, understand it, and then act upon it. Attention and WM exhibit strikingly similar patterns in the cerebral cortex, suggesting that they leverage shared neural mechanisms [6,7]. Top-down attention, also known as voluntary attention, is a goal-directed process, meaning that our expectations, experience, and rewards guide what we attend to. Bottom-up attention, also referred to as reflexive attention, is a stimulus-driven process in which sensory events capture our attention [8]. These two forms of attention, steered by an individual's current behavioral goals and shaped by learned priorities based on personal experience and evolutionary adaptations, have been in competition or cooperation constantly [9].

Attention and WM are expected to equip bionic robots with multimodal information processing and internal state guidance capabilities, thereby compensating for the limitations of existing memory architectures in enhanced cognitive learning and executive functions [10]. However, most implementations rely on software algorithms operating within a Von Neumann architecture, where computation and storage are physically segregated, leading to the memory wall problem [11]. More efforts are urgently needed to realize brain-inspired functions and computation simultaneously in an in-memory computing way. Postulated in 1971 [12] and first fabricated in 2008 [13], the memristor, an ideal synapse-like device, whose resistance is regulated by the amount of electrical charge flowing through itself, is proposed as a novel circuit element. Owing to their superior characteristics, such as nanoscale size, non-volatility, resistance plasticity, and compatibility with Complementary Metal-Oxide-Semiconductor (CMOS) [14–16] memristors are widely used in bionic circuits.

Proposed memristive circuits simulated neuronal and synaptic features and enabled various bio-inspired perceptual and cognitive functions, including sensory processing, emotional evolution, associative memory, and decision-making [17–21]. Jiang et al. [22] distinguished and implemented three types of memory in a memristive bionic memory circuit, where variable learning and forgetting rates were realized, considering stimulation intensity and attention mechanisms. In [23], an array of planar 2D channel memristors was designed, and the information encoding, forgetting, retrieval, and updating functions of WM were realized. A memristive cross-modal associative memory circuit was proposed in [24], and the interaction of information processing, such as synergy, facilitation and transmission during olfactory and visual conditioning was realized. Deng et al. [11] constructed a memristive neural network circuit to achieve learning and decision-making functions. In addition, based on the internal state like experience memory, an OC-CC cascaded circuit was proposed in [25], which enabled the circuit to continuously reshape its own memories and actions. Sun et al. [26] achieved emotional habituation with contextual dependency. Based on Hull's secondary learning system, factors that influence decision-making, such as demand states, incentive motivation, and habit strength, were considered in [27].

Consequently, research focuses on addressing real-time cognitive challenges and decision-making tasks, but is unable to deal with noisy, weak, or ambiguous perceptions and goal changes within dynamic environments. On one front, the presented research concentrates on the stimulus-association-behavior paradigm, using brain-inspired mechanisms such as associative memory, different types of human memory, and the integration of multimodal and cross-modal information [18,20,22,28]. On another front, significant strides have been made in simulating the OC-CC learning process and exploring the driving and regulating processes of internal states, such as habitual memory, expectation, experience, and emotional evolution [19,25–27]. However, despite the fact

that external multimodal stimuli and internal states are two of the most critical information pathways for the biological brain, few studies have successfully integrated them into a circuit. This separation fails to mimic how humans utilize perception, knowledge, internal states, and goals for advanced cognition and effective action [29].

Therefore, in this work, to create a framework that seamlessly selects relevant information from both the external environment and internal regulation, an attempt is made to unify working memory and the two forms of attention. We construct an attention-regulated working memory model. Based on this model, we design and verify the memristive circuit in LTSPICE to achieve rich cognitive control functions in an on-line in-situ learning and in-memory computing manner. Finally, our circuit can be adaptively applied to the cognitive control and execution system of robots within complicated circumstances and tasks. The main contributions of this work are summarized as follows.

- (1) Inspired by the working memory structure and the top-down and bottom-up attention mechanisms in cognitive neuroscience, we propose an attention-regulated working memory model. This model processes current external multisensory stimuli and retrieves stored knowledge from internal reinforcement simultaneously, leading to a learnable executive action.
- (2) Considering different audio-visual input patterns and varying stimulus salience, the memristive circuit is capable of rich functions, such as selective attention, multimodal filtering and encoding, bias formation, context control, and adaptive decision-making. Moreover, the proposed circuit realizes online in-situ learning and in-memory computing based on memristors, which provides advantages for applications on resource-constrained terminals.
- (3) By regulating coordinative, complementary or competitive interactions among the selections and their two associated attentional pathways, the simulation results demonstrate that our circuit can be adaptively applied to complicated circumstances. It is verified to promptly cope with the noisy, feeble or ambiguous perception, goal changes and context switches.

The rest of this paper is organized as follows. Section 2 proposes the attention-regulated working memory model based on the cognitive neural mechanism and constructs the experimental design. Section 3 delineates the memristive circuit and the corresponding simulation results including phonological loop, visuospatial sketchpad, the attention control module, episodic buffer, and central executive module. Section 4 simulates the proposed circuit in cognitive control and execution tasks, performing the circuit analysis and comparison. Finally, the conclusions are drawn in Section 5.

2. Working memory model and its circuit framework

Working memory is composed of four main components, including the central executive control, and three subsidiary storage systems, namely, phonological loop, visuospatial sketchpad, and episodic buffer. Besides, an attention control system that drives the whole system and deals with cognitive tasks is functionally emphasized [30]. The four parts are responsible for different forms of multimodal information storage and processing, especially visual information and auditory information. They work together to perform a wide range of coding and manipulation activities. Accumulating evidence has established an association between WM and the abilities to remember, maintain attention, and inhibit irrelevant data while switching between tasks. It is a critical cognitive skill that is mainly associated with adaptive strategies, task-switching, decision-making, and directing attention toward salient information, enabling organisms to make rapid and appropriate decisions when engaging with their environment [31,32].

As the two primary modalities within the working memory model, auditory and visual systems are closely linked in the brain, with

their processing centers in proximity, facilitating constant information exchange. Research has found that auditory and visual information can be mutually reinforcing [33]. For instance, when hearing and vision match, children exhibit increased focus, faster learning, improved memory retention, and slower forgetting. When singing, it is easy to remember the lyrics and tunes if you can see a picture that matches the content of the lyrics. The integrated effect of audio-visual stimulation enhances the attention and memory processes in relevant brain regions, surpassing the impact of unimodal stimulation.

Moreover, attention can be value-driven, with reward history exerting a direct influence on selection [34]. Giving rewards multiple times along with a certain action is considered a learning process, mainly held in the brain's Prefrontal Cortex (PFC) [35]: it is here that the associations between rewards and actions are formed and stored, obtaining the top-down attentional bias signals [8,36], which do not disappear over time. This learning process involves accumulating experience and extracting useful information from it. Biasing is possible due to reward signals that reinforce subsequent active selections and current actions, aligning them with the corresponding situation and goal.

The proposed bionic attention-regulated working memory circuit framework is illustrated in Fig. 1. In the bottom-up attentional pathway, for each group of target and bionic WM unit, the auditory and visual stimuli in the environment or from SM are represented correspondingly by analog voltage signals V_{audio} and V_{visual} , serving as the inputs of the circuit. The magnitude of the analog voltage carries the salience of external stimuli. After filtering, synchronizing and encoding in the phonological loop and visuospatial sketchpad, output spike signals V_{en} are generated and transmitted to the attention control module. And through the capture and maintenance of stimulus-driven attention, the output signal V_{att} (bottom-up attentional salience signal), which represents the strength of attention captured by external stimuli, will be delivered to the central executive control module. And in the top-down attentional pathway, the association between reinforcement signals V_{reward} and a certain action V_{out} is formed and stored in the episodic buffer, obtaining the top-down attentional bias signal V_{bias} (the top-down attentional bias signal), which represents the internal knowledge and experience-driven bias. V_{trans} , which indicates a transformation of task or context, can reset V_{bias} . V_{out} derives from the central executive control module and represents the currently being performed action, ultimately issuing PWM-level commands to the motor driver in robots.

It compares signal strengths of V_{att} and V_{bias} which result from the two attentional pathways mentioned above.

When faced with weak or ambiguous bottom-up perceptual stimuli, the bottom-up attentional signal V_{att} is weak, and retrieving bias signals from LTM can significantly enhance the performance of WM [37]. In such complementary cases, decisions are made based on the top-down attentional bias signal V_{bias} from the episodic buffer, which represents the knowledge and task-related experience stored in LTM. When external perceptual stimuli and internally formed bias point toward the same decision, V_{att} and V_{bias} work cooperatively, with execution carried out more swiftly. In contrast to complementarity or cooperation, there is a competitive interaction among the selections and their two associated attentional pathways. In complex and changing environments and tasks, V_{att} essentially contains the environmental externalities that are crucial for an organism's perception and performance. V_{bias} , which is evoked by internal information through learning or reinforcement processes, is not allowed to override the perceptual attention V_{att} and dominate the system. Therefore, V_{att} is given higher priority than V_{bias} .

In applying this circuit to the cognitive control and execution system of robots, the potential behavioral outcomes for the robot are binary: either Action1 or Action2, such as turning left or right. As shown in Fig. 2, the interaction between stimulus-driven bottom-up attention and the top-down bias of reward reinforcement within the framework of WM, and its impact on executive decision-making, is discussed in detail. Specific simulation results of the 10 processes will be analyzed and explained in Section 4.

3. Memristive circuit design for working memory networks

3.1. Memristor model

In this work, the AgInSbTe (AIST) model [38] is adopted to mimic homeostatic synaptic plasticity [39], and is described as

$$v(t) = [R_{on}x + R_{off}(1 - x)]i(t), \quad (1)$$

$$\frac{dx(t)}{dt} = \begin{cases} k_{on}v(t)f(x), & v(t) > V_{on} > 0, \\ 0, & V_{off} \leq v(t) \leq V_{on}, \\ k_{off}v(t)f(x), & v(t) < V_{off} < 0, \end{cases} \quad (2)$$

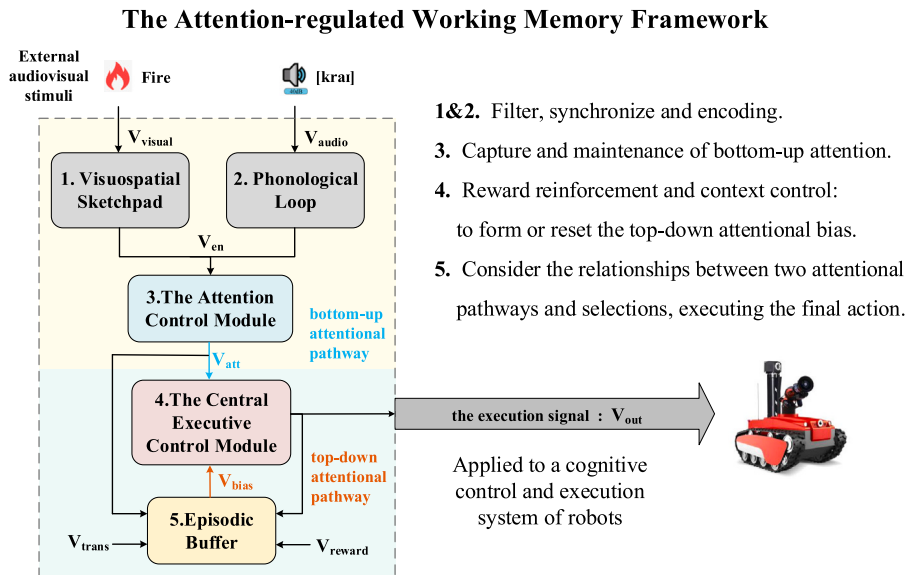


Fig. 1. The proposed bionic attentional-regulated working memory circuit framework for cognitive control and execution. It also illustrates the data flow, attentional pathways and vital functions of each module.

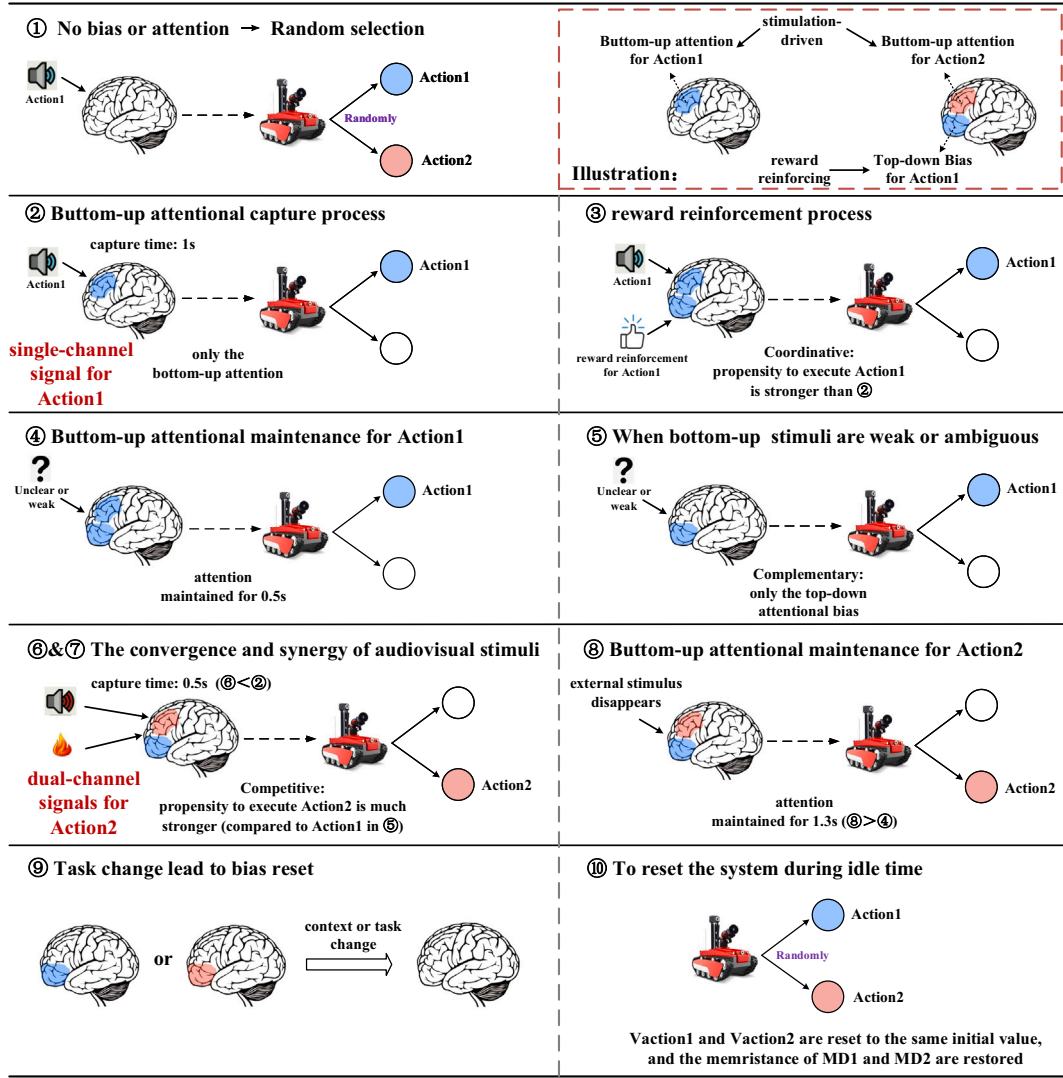


Fig. 2. Describe the interaction (coordinative, complementary or competitive) between stimulus-driven (externally single-channel or dual-channel signals) bottom-up attention and the top-down bias of reward reinforcement and its impact on executive decision-making (Action1 or Action2).

$$f(x) = \begin{cases} (a(1-x))^p, & v(t) > 0, \\ (ax)^p, & v(t) < 0, \end{cases} \quad (3)$$

where $v(t)$ and $i(t)$ represent the voltage and current across the memristor, respectively, and the resistance of memristor, or memristance, is variable. R_{on} denotes the lowest memristance, or rather the low-resistance state (LRS) where the memristor is completely doped, and R_{off} denotes the highest memristance corresponding to high-resistance state (HRS) and completely undoped. The state variable $x(t)$ is a normalized width of the conducting layer, whose derivative is matched to the current voltage $v(t)$. $f(x)$ is a speed adaptive state variable function. V_{on} and V_{off} are positive and negative threshold voltages, respectively. The change of state variable occurs only when the threshold voltage is surpassed. k_{on} , k_{off} , a and p are adjustable fitting parameters that enable the model to accurately describe the behavior of various memristive devices. The parameters of the memristors used in this work are given in Table 1.

The simulation results of memristance change under the varying applied voltage are presented in Fig. 3. Suppose M_0 and M_S indicate the memristance. In Fig. 3(a), voltages with various amplitudes are applied across M_0 and simulations have been carried out to show the three crucial properties of the AIST model: (i) when the applied voltage exceeds

the threshold voltage (± 2 V), the memristance decreases to LRS at positive voltages and increases to HRS at negative voltages; (ii) when the applied voltage does not exceed the threshold voltage, the memristance remains constant; (iii) the memristance varies non-linearly, changing fast at first but slowly at the end of every transition, which is consonant with the characteristics of biological synapses. That is, the stronger the synaptic strength, the slower it will change. Besides, in Fig. 3(b), voltage pulses with the same frequency (50 Hz) and different duty ratios (50 %, 40 %, 25 %) are applied to M_S , where pulses with higher duty ratios

Table 1

The parameters of memristors.

Parameters	M_0/M_S	M_A/M_V	M_C	M_M	M_T	M_R	M_D
$R_{on}(k\Omega)$	5	10	5	5	5	5	5
$R_{off}(k\Omega)$	100	100	100	100	20	20	100
$V_{on}(V)$	2	0.2	0.2	0.2	1.2	0.2	0.2
$V_{off}(V)$	-2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
$k_{on}(A)$	100	3300	800	500	100	50	50
$k_{off}(A)$	100	3300	2000	80	100	50	80
a	0.27	0.2	0.27	0.27	0.27	0.27	0.27
p	1.58	2	1.58	1.58	1.58	1.58	1.58

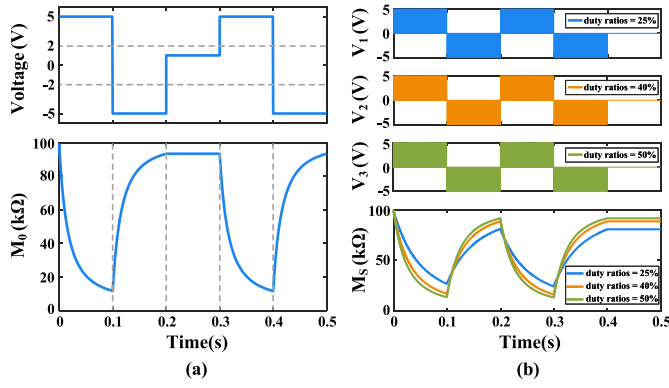


Fig. 3. Simulation results of M_0 and M_S on LTSPICE. (a) The alteration of M_0 when applying positive and negative voltages with different amplitudes. (b) The alteration of M_S when applying voltage pulses whose frequencies are all 50 Hz. Variable duty ratios (or the pulse width) of the applied voltage pulses are leveraged to show their effects.

conduce to faster decrease and increase of M_S . According to Fig. 3, the AIST model is extremely applicable for mimicking information manipulation in the brain, as well as changing the cognitive function status in subsequent modules.

3.2. Phonological loop and visuospatial sketchpad

WM serves as a multimodal information interface, synchronizing and filtering key information from the redundant flow of environmental data [30]. In WM, the phonological loop is responsible for the temporary storage of verbal and acoustic information. Meanwhile, the visuospatial sketchpad operates in parallel to the phonological loop, processing and holding visual or spatial information. This dual-component system allows for the efficient management of multimodal information, ensuring that both auditory and visual data are effectively integrated and utilized [40]. These two subsystems arguably play a pivotal role in selecting which stimuli are encoded into WM [41]. They effectively filter out stimuli known to be irrelevant from the start, especially when such stimuli can be easily discriminated from relevant ones.

As a result, we propose a synchronized encoding scheme that combines rate and temporal coding schemes [42]. This scheme ensures encoding synchronicity for both the phonological loop and the visuospatial sketchpad, mapping external stimuli into characteristic electrical pulses in a specific pattern while filtering out distractions. Accordingly, the phonological loop and visuospatial sketchpad based on memristor are shown in Fig. 4. V_{audio} and V_{visual} are analog voltages in the range of 0-1 V, representing the salience of the external input stimulus. V_{syn} represents the synchronization pattern and is an internal clock signal in the frequency range of 25-100 Hz. Note that V_{syn} in phonological loop and visuospatial sketchpad systems must be the same. Therefore, the phonological loop and visuospatial sketchpad in WM primarily provide three functions: (1) filtering, to filter out input signals with low task relevance or intensity, avoiding interference with subsequent processing and decision-making; (2) synchronizing, to adopt a signal synchronization pattern based on the same frequency, ensuring that WM can act as an interface to connect multimodal inputs; (3) encoding, to generate spikes with millisecond precision closely analogous to biological neurons, which have strong anti-interference capability to change memristance in subsequent modules. These features allow our circuits to analyze consistent or conflicting semantics of audio-visual inputs in the next attention control module, enabling differentiation in the capture and maintenance of attention.

Take the phonological loop for example. N_1 , as the main switch of the module, is used to realize the refractory period of neuronal firing, and it

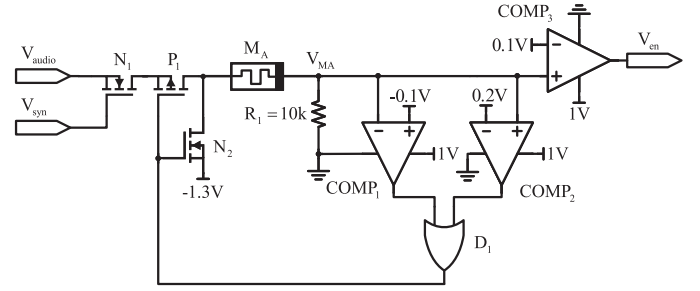


Fig. 4. The circuit diagram of the phonological loop. N_1 and N_2 are IRFH6200 and FDS6930A, whose turn-on voltages are 1 V and 1.5 V. P_1 is FDS4435A and its turn-on voltage is -1 V.

is controlled by the same V_{syn} as the visuospatial sketchpad. When V_{syn} is at a high voltage of 1.5 V, N_1 is turned on, so the neuron enters the firing process; otherwise, N_1 is turned off, and the neuron enters the refractory period. At first, M_A is initialized to HRS, and V_{MA} is at the resting potential. During the firing period, when an external multisensory stimulus is input, MOSFETs P_1 and N_1 are turned on, and V_{audio} is applied to the positive terminal of M_A as the input of the circuit. The memristance of M_A gradually decreases at a positive voltage, and owing to the existence of R_1 , the voltage at V_{MA} gradually increases. When V_{MA} increases to more than 0.1 V, $COMP_3$ outputs high voltage as V_{en} . The rising delay of the output signal V_{en} increases as the amplitude of V_{audio} decreases, while the pulse width decreases accordingly, that is, the neuronal firing time decreases, which is due to the different memristance change rates of the memristor at different applied voltages. During the refractory period, or when V_{MA} keeps going up to more than 0.2 V, voltage comparator $COMP_2$ and OR gate D_1 output high voltage, causing P_1 to turn off and N_2 to turn on. Since then, a reset voltage of -1.3 V is applied to the M_A , V_{MA} jumps to negative and is less than -0.1 V, so $COMP_1$ outputs low and $COMP_2$ outputs high. As M_A gradually increases under negative voltage, V_{MA} gradually increases until it exceeds -0.1 V, and $COMP_1$ and $COMP_2$ both output low, which makes D_1 output low. When V_{syn} goes up to 1.5 V again, V_{audio} is applied to M_A and repeats the above firing process, outputting V_{en} . Since the rising rate of V_{MA} depends on V_{audio} and V_{syn} , V_{audio} is used to control the neuronal firing time and the same V_{syn} is used to ensure the audio-visual encoding synchronicity. Similarly, the design of the visuospatial sketchpad operates on the same principles. It employs components such as N_3 , N_4 , P_2 , M_V , $COMP_4$ - $COMP_6$, and D_2 to achieve its functionality. The simulation results of the phonological loop are shown in Fig. 5.

3.3. The attention control module

When driven by significant multimodal stimuli that are task-related and semantically consistent, the attention control module can capture our attention at the fastest speed and for longer periods of time. When the significance of the stimulus decreases, the capture becomes slower and the maintenance time becomes shorter [32]. Based on the biological mechanisms above, attentional capture module and attentional maintenance module are responsible for mimicking the capture of multisensory stimulation-driven attention, and self-maintenance after the disappearance of stimuli, respectively. As illustrated in Fig. 6, the attentional capture module judges the intensity and pulse width of spike signals V_{en1} from the phonological loop and V_{en2} from the visuospatial sketchpad. It then converts these signals into an intermediate signal, V_{cap} , which regulates the input and determines the maintenance time of the subsequent attentional maintenance module. The output signal V_{att} will be transmitted to the central executive control module, where different V_{att} reflect how the brain processes and integrates information from different senses and how it adjusts attention resources in the face of

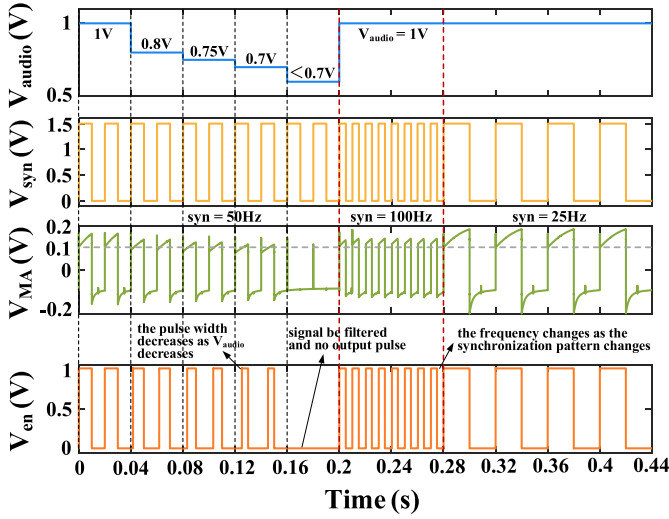


Fig. 5. The simulation results of the phonological loop. Between 0 and 0.2 s, keep the same synchronization pattern (50 Hz), and different amplitude of the multi-sensory inputs V_{audio} (1 V, 0.8 V, 0.75 V, 0.7 V, 0.6 V) are applied. Respectively, the rising delay of the output signal V_{en} increases as the amplitude of V_{audio} decreases, the pulse width decreases accordingly and the neuronal firing time also decreases. When $V_{audio} < 0.7$ V, the signal will be filtered out and no output pulse will be generated. Between 0.2 s and 0.44 s, V_{audio} is maintained at 1 V and different synchronization patterns V_{syn} (100 Hz, 25 Hz) are applied. Respectively, the frequency of V_{en} changes as the synchronization pattern changes.

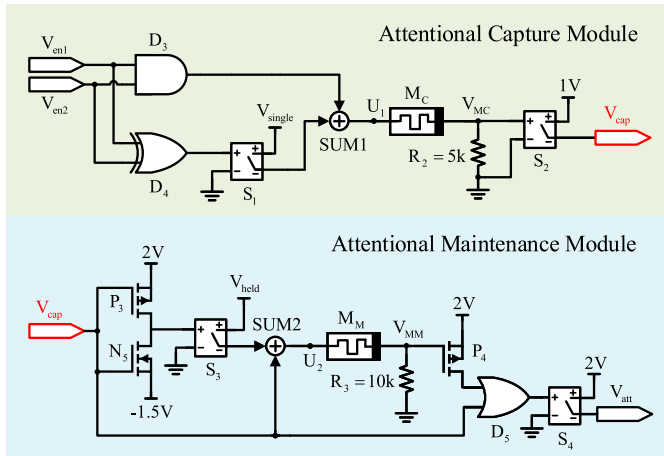


Fig. 6. The circuit diagram of the attention control module consists of two parts to achieve the capture of multisensory stimulation-driven attention, and self-maintenance after the disappearance of stimuli.

different cognitive demands. The two circuits are elaborately depicted as follows:

- (1) **Attentional Capture Module:** When the single-channel (encoded auditory or visual) signal is input, the encoded pulse voltage is fed into XOR gate D_4 . D_4 outputs a high-level signal to turn on the voltage-controlled switch S_1 . Due to the conduction of S_1 , a pulse signal V_{single} , whose frequency and amplitude are the same as V_{en1} and V_{en2} , but whose pulse width (5 ms) is always lower than both, is applied to the positive terminal of M_C through SUM_1 . When the dual-channel auditory and visual voltage signals are input simultaneously, the output signal of AND gate D_3 is the one with the smaller pulse width in V_{en1} and V_{en2} . Due to D_3 , D_4 and

SUM_1 , different input patterns and varying stimulus salience result in different U_1 . Then U_1 is applied at the positive terminal of M_C , which causes different rates of decrease in the resistance of M_C . When $V_{MC} > 0.43$ V, the voltage-controlled switch S_2 is conducted, and 1 V voltage is output to V_{cap} .

- (2) **Attentional Maintenance Module:** the positive voltage V_{cap} is applied to M_M to reduce its resistance first, and V_{cap} can be reversed by P_3 and N_5 . When the input signal exceeds the switching threshold of N_5 , a signal of -1.5 V is output, otherwise a signal of 2 V is output. Then through the voltage control switch S_3 , when the signal input disappears, a negative pulse signal V_{held} is applied to M_M , to reset its resistance. During recovery of M_M , V_{MM} gradually rises from a negative value until it exceeds the threshold of -0.1 V, causing the output V_{att} to become 0. This transition marks the end of the maintenance time, defined as the interval from the disappearance of the input signal V_{cap} until V_{att} reaches 0. Therefore, the maintenance time primarily depends on the recovery speed of M_M from LRS to HRS. The maintenance time can be changed by adjusting the parameters of M_M and changing the V_{held} . Specifically, increasing R_{off} from 100 k Ω to 150 k Ω can extend the maintenance time by delaying charge dissipation, and increasing $|V_{held}|$ (changing V_{held} from -1.5 V to -3 V) can shorten maintenance time by accelerating the recovery. Finally, the two modules are connected through the OR gate D_5 , and then through the voltage-controlled switch S_4 , the output V_{att} is 2 V.

The simulation results are shown in Fig. 7. In (a)–(e) the attentional capture is realized, and in (e)–(g) the attentional maintenance is realized. Take the case of the first column (0–3 s) with only visual input as an example. During 0–1.5 s, V_{en1} with 0 V and V_{en2} with a pulse width of 10 ms are input. The voltage at U_1 is V_{single} with the smallest pulse width (5 ms), resulting in the slowest decrease in M_C . Due to the presence of R_2 , V_M rises the slowest and reaches the threshold of 0.43 V at the latest. This causes V_{cap} to output at 1 s–1.5 s and is applied to M_M so that its resistance decreases. After 1.5 s, the input ends, V_{held} makes the resistance of M_M gradually rise, and the output V_{MM} turns over to a negative voltage. Until $V_{MM} > -0.1$ V, the output signal V_{att} is 0. Thus, V_{cap} is maintained during the recovery time of M_M and V_{MM} within 1.5 s–2 s. Finally, V_{att} outputs 2 V during 1 s–2 s, indicating that under single-channel stimulation, the brain processes sensory information slowly, with the bottom-up attention captured slowly and maintained for a short time. The second column (4 s–7 s) shows the significant inputs of audio-visual dual-channel. The pulse widths of V_{en1} and V_{en2} are both 10 ms, and V_{att} outputs in 4.5 s–6.8 s, indicating that under the convergence and coordination of audio-visual stimuli, attention and information processing in relevant brain regions are stronger than those of single-channel, with attention captured the fastest and maintained for a long time. In the third column (8 s–11 s), the pulse width of V_{en1} is 8 ms, which means the significance of auditory stimuli is low, and the task relevance and semantic consistency are weaker than those in the second column. V_{att} outputs in 8.6 s–10.1 s, which is between the first two cases.

3.4. Episodic buffer

As we can see, WM will maintain for a limited period and eventually disappear unless continuously applying task-related inputs. Unlike WM, information in PFC is permanently maintained and can be retrieved into the episodic buffer and then into the WM system as the top-down attentional bias signal [43]. The bias signal is only reversed or reset when the context or task changes. According to the mechanisms above, the circuit implementation of the episodic buffer has been represented in Fig. 8, which includes the context control module and reward reinforcement module, and correspondingly realizes the following two functions:

- (1) **Context Control Module:** The context signal V_{trans} indicates a transformation of a task or situation, or can be applied to the module

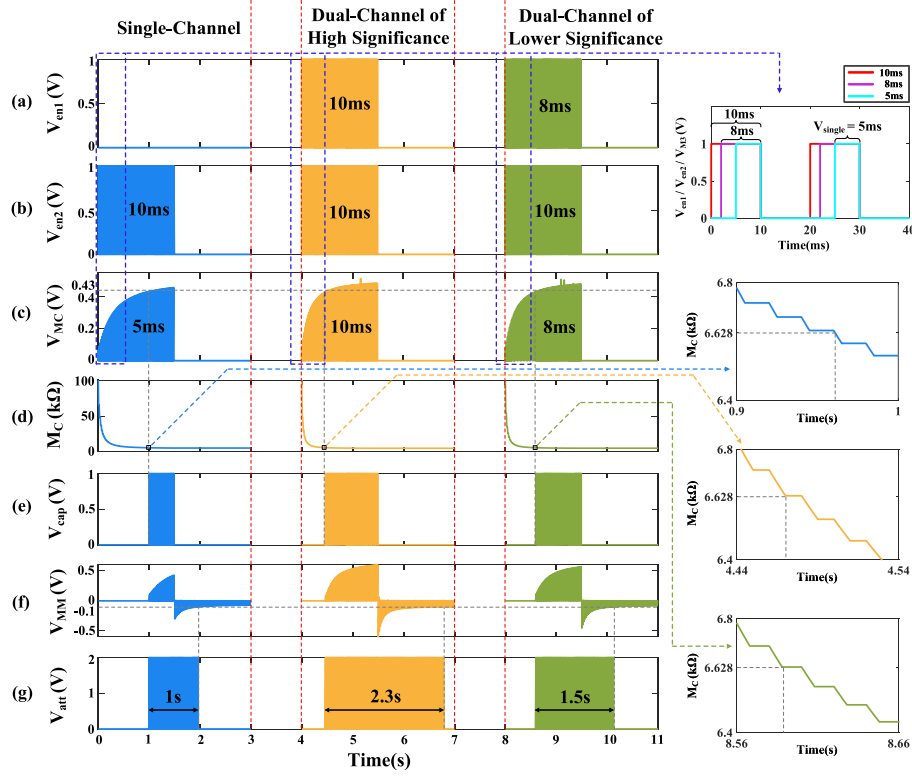


Fig. 7. Simulation results of the attention control module. The first column (0–3 s) shows the case with only visual input. The second column (4–7 s) is the case of significant input of audio-visual dual channels. In the third column (8–11 s), the significance of auditory stimuli is low, and the task relevance and semantic consistency are weaker than that in the second column. All three have inputs only for the first 1.5 s. M_C and V_{MC} distinguish the capture speed under different stimulus combinations and the recovery time of V_{MM} realizes the self-maintenance of attention. The duration of the output V_{att} reflects the difference between the capture and maintenance of attention.

after multiple errors. Then it is converted into an intermediate signal V_{update} , as a reset signal input into the following reward reinforcement module. The components Div_1 and Div_2 are dividers with a minus sign, that is, $V_{div1} = -IN_1/IN_2 = M_T/R_{f1}$, so V_{div1} is proportional to the resistance of M_T . The change in the context signal V_{trans} from 0 to 2 V indicates that the contextual information

has changed. The memristance of M_T decreases because the amplitude of V_{trans} exceeds the positive threshold voltage of M_T . With the decrease in the value of M_T , the output of Div_1 is also reduced. The comparator is composed of $COMP_7$, $COMP_8$, and D_6 . If $0.55 \text{ V} < V_{div1} < 0.7 \text{ V}$, the comparator will output a high level and input into OP_2 . During this period, reset signal V_{update} will be generated with an amplitude of -1.5 V . When V_{trans} exceeds the threshold voltage of M_T , the reset signal V_{update} is applied to M_R to increase the value of M_R , resulting in the formed bias signal V_{bias} being reset. Therefore, when contextual information changes, the result of reward reinforcement will be affected.

- (2) **Reward Reinforcement Module:** V_{reward} serves as a reinforcement signal, V_{update} serves as a context change signal and V_{out} derives from the central executive control module and represents the currently being performed action. When V_{reward} is given several times during an action, the association between the reward and the behavior V_{out} is created and stored, and the top-down attention bias signal V_{bias} is formed, which does not disappear over time, achieving the accumulation of experience and extracting useful information from the experience. The memristance of M_R is inversely proportional to the strength of the association between reward and behavior and is initially at HRS. The voltage U_3 at the positive terminal of M_R is controlled by D_7 and SUM_3 , together with the input signals. When neither D_7 nor V_{update} has input, U_3 is a read voltage of 0.5 V , which is below the positive threshold voltage of M_R and keeps M_R unchanged. When both V_{out} and V_{reward} are at high levels, it represents the learning process. The AND gate D_7 outputs a high level and U_3 is 1.5 V , which exceeds the positive threshold voltage of M_R , so the memristance of M_R decreases. Moreover, when the context

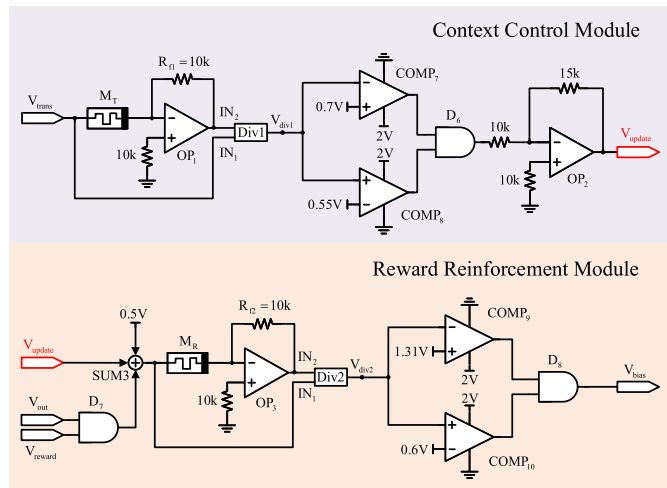


Fig. 8. The circuit diagram of the episodic buffer consists of two parts to achieve reward reinforcement, forming the top-down attentional signal V_{bias} . When the contextual information V_{trans} changes, V_{update} is generated to influence V_{bias} .

control module outputs V_{update} , representing the reset process, U_3 is -1 V, exceeding the negative threshold voltage of M_R , so the memristance of M_R increases. With the change in the value of M_R , $V_{div2} = -IN_1/IN_2 = M_R/R_{f2}$, so the output of Div_2 is also changed. The comparator is composed of $COMP_9$, $COMP_{10}$, and D_8 . If $0.5 \text{ V} < V_{div2} < 1 \text{ V}$, the comparator will output a high level of V_{bias} .

The simulation results are shown in Fig. 9. In (c)–(e) the context control is realized, and in (a)–(b) and (f)–(g) the reward reinforcement is realized. In the reinforcing stage (0–0.8 s), V_{out} and V_{reward} are presented at the same time, and when V_{div2} is below 1 V, the response V_{bias} is triggered at 0.46 s. The decrease in the M_R and V_{div2} indicates the accumulation of experience which reinforces subsequently active selections and current actions. On the other hand, in the context control stage (0.8–2 s), the context signal V_{trans} (2 V) appears to indicate the transformation of the task or situation. During 1.3 s–1.6 s, as M_T decreases, V_{div1} drops to a certain voltage range (0.55 V–0.7 V), triggering the intermediate signal V_{update} (-1.5 V), which can raise M_R as a reset signal. Therefore, when V_{div2} returns above 1 V (about 1.37 s), V_{bias} is reset to 0. In particular, during 0.8 s–1.3 s, M_R and V_{div2} remain the same, indicating that the information in the PFC is permanently preserved. That is, the formed bias does not disappear over time, similar to the LTM.

3.5. The central executive control module

Research has revealed that the final competition occurs in the central executive module. It compares signal strengths that result from the combined effects of the quality of the encoded multisensory inputs, top-down bias signals, and bottom-up attentional salience signals [31,44]. Inspired by the mechanisms above, the circuit design of the central executive module is demonstrated in Fig. 10(a), where multimodal inputs from SM are processed, and stored knowledge from LTM is retrieved. In our design, for each selection, there are two pathways of attention. V_{att} from

the attention control module and V_{bias} from the episodic buffer serve as the inputs to this module, where the cooperative or competitive relationships of two attentional pathways are considered, thus transforming from a random exploration process to the final learnable decision and action. The basic functions of the central executive module are explained as follows:

The switching parameters of memristors exhibit statistical fluctuations from cycle to cycle, which enable true random number generation and facilitate random exploration [45]. At the inception of the demonstration, stimulus V_{act} (0.5 V) can randomly trigger either Action1 or Action2. U_4 , the voltage at the positive terminal of M_{D1} is the sum of V_{act} , V_{att1} and V_{bias1} corresponding to Action1. U_5 , the voltage at the positive terminal of M_{D2} is the sum of V_{act} , V_{att2} and V_{bias2} corresponding to Action2. When the sum signals (U_4 and U_5) are absent or equal, a random exploration process is initiated based on the intrinsic variability of M_{D1} and M_{D2} , leading to a decrease in the memristance of these two memristors. Specifically, when $V_{action1}$ is less than $V_{action2}$, the output signal of $COMP_{11}$ is positive and switch S_5 is conductive, causing V_{out} to output a $+5$ V signal, indicating that Action1 is selected for execution. On the contrary, when $V_{action2}$ is less than $V_{action1}$ and V_{out} outputs a -5 V signal, and Action2 is executed. During idle time, V_{act} outputs a reset voltage to restore the memristance of M_{D1} and M_{D2} . The voltage of the execution signal V_{out} , as illustrated in Fig. 10(b), is primarily influenced by the difference between U_4 and U_5 .

3.6. Complete circuit

The complete circuit of the attention-regulated working memory system, as depicted in Fig. 11, consists of two sub-circuits (the upper sub-circuit is associated with Action1 and the lower sub-circuit is associated with Action2) and a central executive control module. The sub-circuits comprise the phonological loop and visuospatial sketchpad, an attention control module, and an episodic buffer. The signal V_{out} , indicated in

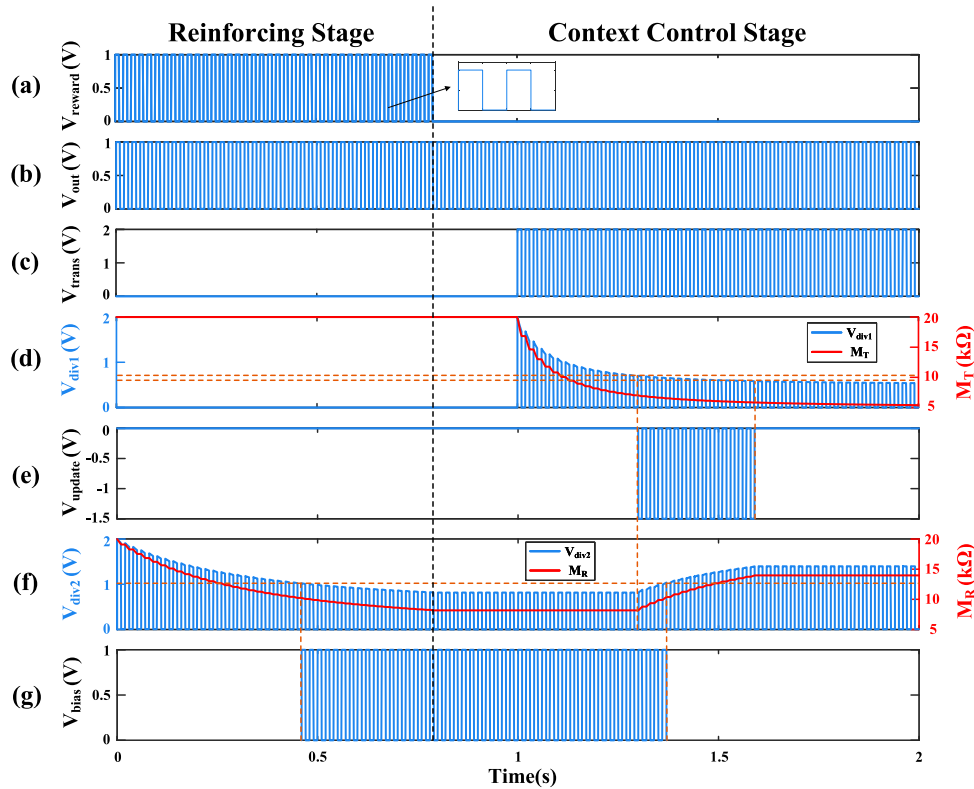


Fig. 9. Simulation results of the episodic buffer. When the context or task changes, the intermediate signal V_{update} can reset formed V_{bias} . The decrease of the M_R and V_{div2} indicates the accumulation of experience in reinforcing stage.

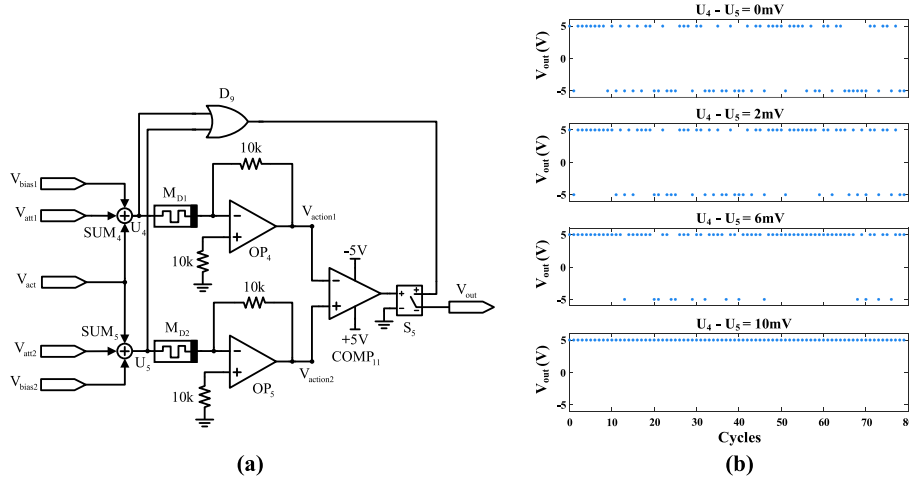


Fig. 10. (a) The circuit schematic of the central executive module. (b) The execution signal V_{out} versus the difference between the sums (U_4 and U_5) of the signals from the two attention channels. Here, U_5 is fixed at 0 mV, and U_4 takes the incremental voltages (0 mV, 2 mV, 6 mV, 10 mV). Over 80 test cycles, as U_4 changes from 0 mV up to 10 mV, the likelihood of V_{out} reaching +5 V gradually climbs, starting from around a 50 % probability when $U_4 = 0$ mV, and rising to nearly 100 % by the time U_4 hits 10 mV. When U_4 exceeds 10 mV, V_{out} can take an unambiguous voltage +5 V per cycle.

red, serves as the output of the entire circuit and is fed back into the episodic buffers of both sub-circuits, where it is reinforced by the reinforcement signal V_{reward} . Moreover, both sub-circuits have common signals, including V_{trans} , V_{reward} , V_{single} and V_{held} .

4. Simulation results and analysis

4.1. Simulation results of the cognitive control and execution

As illustrated in Fig. 12, V_{audio} and V_{visual} are the inputs of the circuit, representing external auditory and visual stimuli. V_{audio1} and $V_{visual1}$ represent the salience of stimuli under the target of Action1, while V_{audio2} and $V_{visual2}$ represent the salience of stimuli under Action2. These can achieve various synchronization patterns (controlled by V_{syn1} and V_{syn2}) to modulate the priority of the two targets. Each selection has both bottom-up and top-down attentional pathways, denoted by V_{att1} , V_{bias1} , and V_{att2} , V_{bias2} , respectively. The signal V_{out} , serves as the output of the entire circuit. A positive voltage of V_{out} signifies the execution of Action1, whereas a negative voltage denotes the execution of Action2. For instance, in complex rescue situations, define Action1 as proactive rescue mode (e.g., approaching survivors, delivering supplies) and Action2 as emergency avoidance mode (e.g., evading obstacles, emergency retreat). In this regard, the PWM commands corresponding to Action1 are to drive the robot to carry out rescue at a faster speed, while Action2 corresponds to triggering the reverse thruster for emergency evacuation. Signals transmitted among different modules are digital, endowing our circuit with robust anti-interference capabilities. The whole process can be elaborately explained as follows, and each process corresponds one-to-one with those in Figs. 2 and 12:

- (1) *Process 1–2*: When the single-channel signal for Action1 ($V_{audio1} = 1$ V and $V_{visual1} = 0$) is input, the bottom-up attention V_{att1} is captured slowly at 1 s. Consequently, during *Process 1* (0–1 s), due to the absence of V_{att} and V_{bias} , V_{act} drives the robotic system to engage in a random selection behavior. Subsequently, in *Process 2* (1 s–2 s), the emergence of V_{att1} (5 ms) leads to a decrement in $V_{action1}$, thereby facilitating the execution of Action1.
- (2) *Process 3*: When executing Action1, V_{reward} is applied to the episodic buffer (2 s–3 s), which reinforces the association between subsequent proactive selections and the currently rewarded Action1. This reinforcement leads to the formation of the top-down attentional bias signal V_{bias1} at 2.37 s. After 2.37 s, with both external perceptual stimuli and internally stored biases converging

on the same decision, the coordination of V_{att1} and V_{bias1} results in a more rapid decrease in $V_{action1}$ (compared to *Process 2*). This indicates that when the two attentional pathways are working in unison, the robotic system's propensity to execute Action1 is stronger, with more focused attention and faster decision-making.

- (3) *Process 4*: V_{audio1} drops to 0.5 V at 3 s, and is filtered out in the phonological loop, that is, the external stimulus disappears. However, due to the self-maintenance of attention, V_{att1} is maintained until 3.5 s. Therefore, during this period (3 s–3.5 s), the conditions of $V_{action1}$ and V_{out} remain consistent with those between 2.37 s and 3 s.
- (4) *Process 5*: The established V_{bias1} does not dissipate with time. And when the bottom-up perceptual stimuli are noisy, feeble or ambiguous, decisions are made by retrieving top-down signal V_{bias} from the episodic buffer. In such complementary cases (3.5 s–5 s), it is necessary to utilize the knowledge and task-related experience stored in LTM to make better choices.
- (5) *Process 6*: When the significant dual-channel signals for Action2 ($V_{audio2} = 1$ V and $V_{visual2} = 1$ V) are input, V_{att2} is captured faster using only 0.5 s (compared to 1 s in *Process 1*). This demonstrates that when audio-visual stimuli are converged and synergistic, the attentional focus and information processing capabilities are more robust compared to those evoked by the single-channel signal.
- (6) *Process 7*: After 5.5 s, V_{att2} and V_{bias1} compete with each other in the central executive module, resulting in $V_{action2}$ being lower than $V_{action1}$, and V_{out} outputs –5 V, driving the robot to execute Action2. In addition, $V_{action2}$ declines much faster (compared to $V_{action1}$ in *Process 2*), which validates that V_{att} has a higher priority than V_{bias} .
- (7) *Process 8*: Similar to *Process 4*, V_{att2} persists for 1.3 s after the dual-channel stimuli have disappeared (compared to 0.5 s in *Process 4*).
- (8) *Process 9*: If goals change, or if the context's changing demands an alternative course of action, the robot must be able to switch from one plan to another. To verify such conditional reset function, V_{trans} applied in the episodic buffer is increased from 1 V to 2 V (8.5 s–8.9 s), and the established V_{bias} will be reset.
- (9) *Process 10*: To make subsequent new decisions, the system must be able to reset $V_{action1}$ and $V_{action2}$ to the same initial values. Thus, during idle time, V_{act} applies a reset voltage (–1 V) to restore the memristance of M_{D1} and M_{D2} , and the robot returns to the random exploration process.

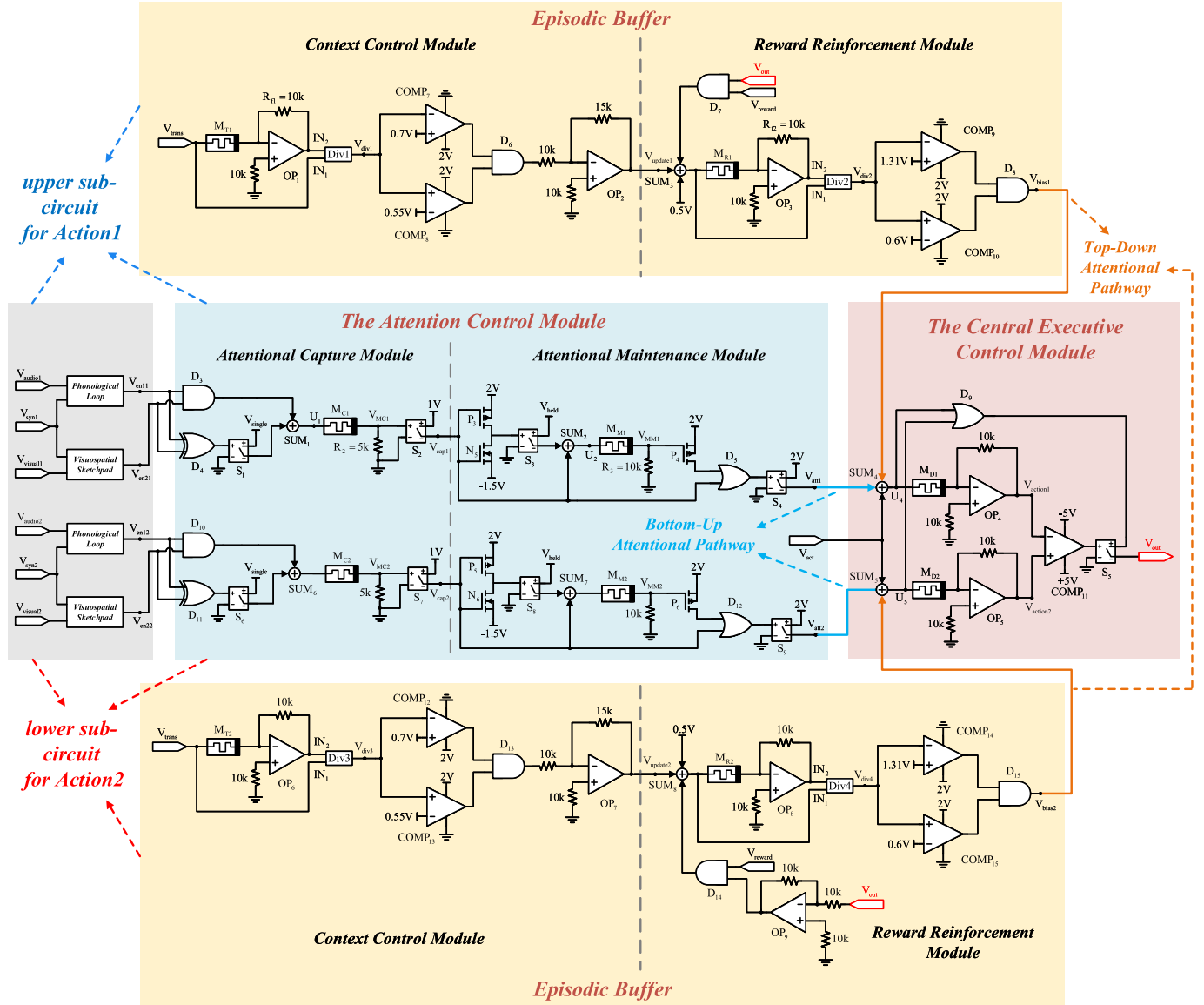


Fig. 11. The complete circuit of the attention-regulated working memory system. P_2/P_4 and P_3/P_5 are FDS4685 and AONR21307, whose turn-on voltages are -1.8 V and -2.1 V. N_3/N_4 is BSC011N03LS and its turn-on voltage is 2.3 V.

4.2. Circuit analysis

In the practical analog circuit, the memristor generates non-ideal deviations in memristance due to conductance drift or device aging, and such deviations include device to device, and cycle to cycle variability, which may affect circuit performance, especially V_{en} , the output of the phonological loop and visuospatial sketchpad. As shown in Fig. 5, the input V_{audio} is 1 V and V_{syn} is 50 Hz. To test the robustness and stability of the encoding pattern brought by the phonological loop, variations of R_{on} , R_{off} , a and p are set to -10% to $+10\%$ using parameter sweeping in LTSPICE, as shown in Fig. 13(a)–(c). Although parameter variations bring slight differences in the decreasing rate of M_A , it can be observed in Fig. 13(d) that V_{en} remains unchanged, which means that slight variation of M_A will not affect the performance of our circuit. Therefore, the excellent robustness of our phonological loop and visuospatial sketchpad is verified.

The operational amplifier in this circuit contains eight MOSFETs and two 5 k Ω resistors. The SUMs consist of an operational amplifier, four 20 k Ω resistors, and one 10 k Ω resistor. The voltage-controlled switches are composed of one voltage comparator and two transmission gates.

Each transmission gate has four MOSFETs. In general, the area of the memristor is about $0.01 \mu\text{m}^2$. A sheet resistance of $25 \Omega/\text{square}$ (poly width is $0.6 \mu\text{m}$) is used to construct resistors under 3 k Ω , while a sheet resistance of $1.2 \text{ k}\Omega/\text{square}$ is used for the resistors over 3 k Ω . In AMI $0.5 \mu\text{m}$ CMOS technology, the channel width and length of the MOSFET are $1.5 \mu\text{m}$ and $0.6 \mu\text{m}$, respectively. The power consumption of this circuit is mainly due to operational amplifiers, resistors and memristors. After a simple calculation, the total power consumption of this circuit is about 18.001 mW. According to the above analysis, the area overheads and average power consumption of the proposed circuits are listed in Table 2.

4.3. Comparison and discussion

Table 3 compares related works in terms of biological fundamentals and functions. [8] has only considered the following four processes of attention in theory: working memory, top-down sensitivity control, competitive selection, and automatic bottom-up filtering for salient stimuli, but these processes have not yet been implemented in circuits. A subject-controlled and variable attention signal is introduced in [22],

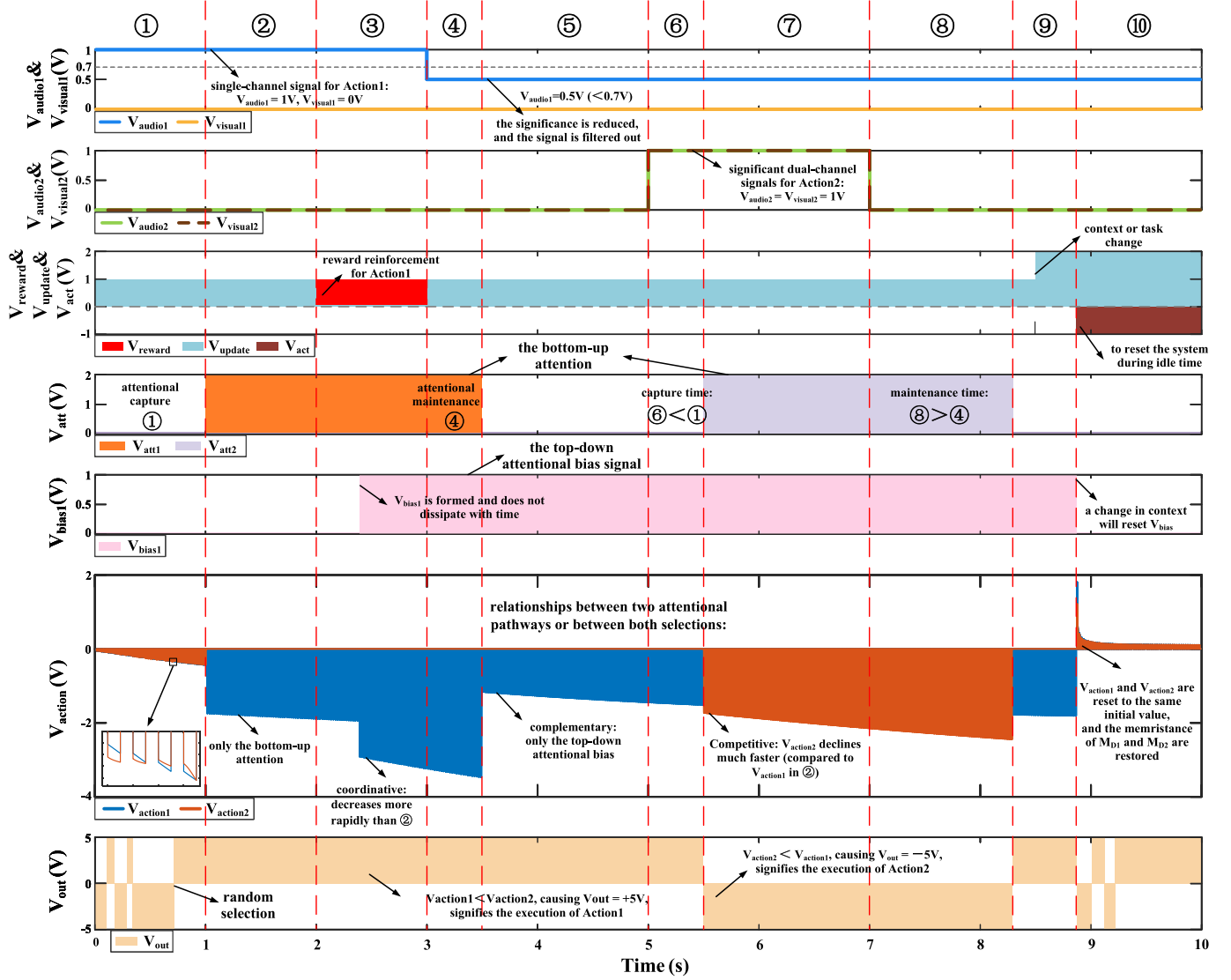


Fig. 12. The circuit simulation results correspond to each process, demonstrating that through competition, collaboration, or complementary between the two attention pathways (V_{att} and V_{bias}) and between both selections (Action1 and Action2), the system can achieve audio-visual filtering and integration, selective attention, reward reinforcement, random exploration and conditional resetting, thereby adapting to target changes and context switches.

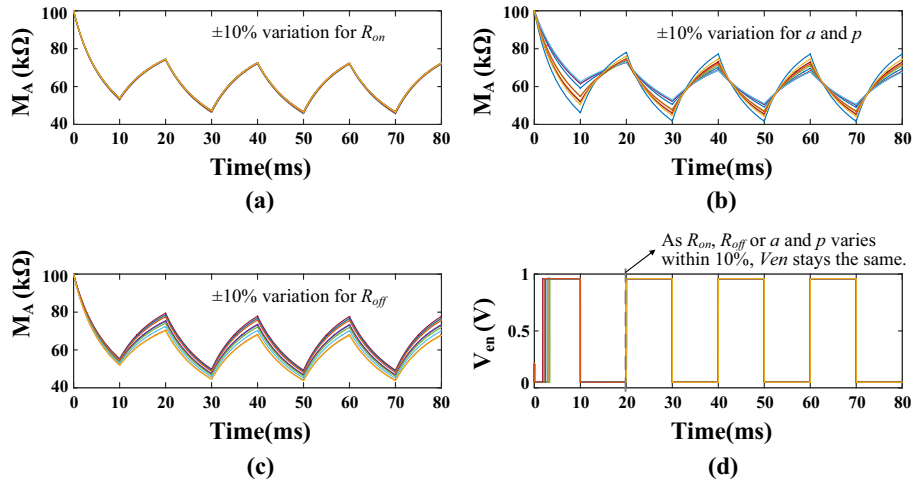


Fig. 13. The variation of M_A from -10% to $+10\%$. The output of the phonological loop and visuospatial sketchpad remains the same.

Table 2

On-chip area estimation and average power consumption of the proposed circuits.

Circuit Module	On-Chip Area	Average Power
Phonological Loop	41.91 μm^2	2.535 mW
Attention Control Module	204.42 μm^2	1.142 mW
Episodic Buffer	138.32 μm^2	1.958 mW
Central Executive Control Module	182.02 μm^2	1.661 mW
Total Circuit	1035.14 μm^2	18.001 mW

Table 3

Comparison of the proposed circuit with other related works.

Works	Biological fundamental	Multimodal processing	Internal states	Functions				Circuit
				Encoding and integration	Selective attention	Context control	Decision-making	
[8]	Attention mechanism in cognitive neuroscience	✓	✓	×	✓	×	×	×
[22]	Three types of human memory	✓	×	✓	×	×	×	✓
[23]	Working memory and hierarchical memory	✓	×	✓	×	×	×	✓
[24]	Associative memory in drosophila	✓	×	×	×	×	×	✓
[25]	Classical conditioning and operant conditioning	×	✓	×	×	×	✓	✓
[26]	Emotional habituation and contextual dependency	×	✓	×	×	✓	×	✓
[27]	Hull's secondary learning system	×	✓	×	×	×	✓	✓
This work	Attention-regulated working memory	✓	✓	✓	✓	✓	✓	✓

but the level of attention is artificially prescribed. An artificial working memory based on memristor array is proposed in [23], but it does not address its ability in multiple cognitive functions. The cross-modal synergy and transfer interactions of input stimulus signals with varying intensities are considered in [24], but it does not involve various scenarios such as internal reinforcement and context control. In addition, based on the internal state like experience memory, Yang et al. [25] have implemented random exploration and experience-based decision-making. In another study [27], factors that influence decision-making, such as demand states, incentive motivation, and habit strength, are considered. However, these studies do not involve the integration and interactions of cross-modal or multimodal stimuli. Based on the internal states like emotion and habituation, Sun et al. [26] take into account contextual control and generalization, but they do not further utilize these for cognitive control and decision-making. Moreover, the realized circuits do not take into account both the processing of external multimodal stimuli and the role of internal states simultaneously. Therefore, this work unifies these two factors with the two forms of attention, constructs an attention-regulated working memory model and designs the circuit based on memristors, which implements richer cognitive control functions, including filtration, integration and encoding for audio-visual signals, sensory and perceptual processing, selective attention, reward reinforcement, context control, random exploration, conditional resetting and executive decision-making.

5. Conclusion

In this work, a brain-inspired attention-regulated working memory framework is constructed, and its memristive circuit design is presented. Inspired by the working memory structure, top-down and bottom-up attention mechanisms in cognitive neuroscience, the processing of external multimodal stimuli and the retrieval of internal reinforcement are achieved and integrated simultaneously. Considering different audio-visual input patterns and varying stimulus salience, filtering, synchronizing and encoding of multimodal signals are realized. In addition to executive decision-making, rich functions such as the capture and maintenance of selective attention, bias formation due to reward reinforcement, context control, conditional resetting, and random exploration are integrated, coping with the absence of stimulation, noisy, feeble or ambiguous perception, goals changing and context switching. By regulating interactions among the selections and their two associated attentional pathways, the simulation results demonstrate that our circuit can be more flexibly applied to the cognitive control and execution system of robots within the complicated circumstances and

tasks. Moreover, the proposed circuit realizes online in-situ learning and in-memory computing based on memristors, which provides advantages for applications on resource-constrained terminals. In the future, our circuits will be expanded and optimized to achieve more abundant functions, improve the accuracy and efficiency of executive actions, and adapt to more complex environments.

CRedit authorship contribution statement

Jihong Zhang: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Xiaoping Wang:** Supervision, Funding acquisition. **Zhanfei Chen:** Supervision. **Chao Yang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the [National Natural Science Foundation of China](#) under Grants 62236005, the Interdisciplinary Research Program of [Hust](#) (Grant No: 2024JCY1006), the [Foundation for Outstanding Research Groups of Hubei Province of China](#) under Grant 2025AFA012, and the 111 Project on Computational Intelligence and Intelligent Control under Grant B18024.

Data availability

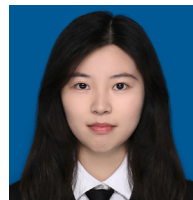
No data were used for the research described in the article.

References

- [1] M. Persiani, A.M. Franchi, G. Gini, A working memory model improves cognitive control in agents and robots, *Cogn. Syst. Res.* 51 (2018) 1–13, <https://doi.org/10.1016/j.cogsys.2018.04.014>.
- [2] L. Davachi, J.P. Mitchell, A.D. Wagner, Multiple routes to memory: distinct medial temporal lobe processes build item and source memories, *Proc. Natl. Acad. Sci.* 100 (4) (2003) 2157–2162, <https://doi.org/10.1073/pnas.0337195100>.
- [3] K.J. Malmberg, J.G.W. Raaijmakers, R.M. Shiffrin, 50 years of research sparked by Atkinson and Shiffrin (1968), *Mem. Cogn.* 47 (4) (2019) 561–574, <https://doi.org/10.3758/s13421-019-00896-7>.
- [4] W.J. Chai, A.I.A. Hamid, J.M. Abdullah, Working memory from the psychological and neurosciences perspectives: a review, *Front. Psychol.* 9 (2018) 401, <https://doi.org/10.3389/fpsyg.2018.00401>.
- [5] N. Hakim, T. Feldmann-Wüstefeld, E. Awh, E.K. Vogel, Controlling the flow of distracting information in working memory, *Cereb. Cortex* 31 (7) (2021) 3323–3337, <https://doi.org/10.1093/cercor/bhab013>.

- [6] Y. Zhou, C. Curtis, K. Sreenivasan, D. Fougner, Common neural mechanisms control attention and working memory, *J. Neurosci.* 42 (37) (2022) 7110–7120, <https://doi.org/10.1523/JNEUROSCI.0443-22.2022>.
- [7] J. Jung, M.A.L. Ralph, R.L. Jackson, Subregions of dlPFC display graded yet distinct structural and functional connectivity, *J. Neurosci.* 42 (15) (2022) 3241–3252.
- [8] E.I. Knudsen, Fundamental components of attention, *Annu. Rev. Neurosci.* 30 (1) (2007) 57–78, <https://doi.org/10.1146/annurev.neuro.30.051606.094256>.
- [9] F. Katsuki, C. Constantinidis, Bottom-up and top-down attention: different processes and overlapping neural systems, *Neuroscientist* 20 (5) (2014) 509–521, <https://doi.org/10.1177/1073858413514136>.
- [10] H. Tan, Y. Zhou, Q. Tao, J. Rosen, S. van Dijken, Bioinspired multisensory neural network with crossmodal integration and recognition, *Nat. Commun.* 12 (1) (2021) 1120, <https://doi.org/10.1038/s41467-021-21404-z>.
- [11] Z. Deng, C. Wang, H. Lin, Y. Sun, A memristive spiking neural network circuit with selective supervised attention algorithm, *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* 42 (8) (2023) 2604–2617, <https://doi.org/10.1109/TCAD.2022.3228896>.
- [12] L. Chua, Memristor-the missing circuit element, *IEEE Trans. Circ. Theory* 18 (5) (1971) 507–519, <https://doi.org/10.1109/TCT.1971.1083337>.
- [13] D.B. Strukov, G.S. Snider, D.R. Stewart, R.S. Williams, The missing memristor found, *Nature* 453 (7191) (2008) 80–83, <https://doi.org/10.1038/nature06932>.
- [14] G. Zhou, Z. Wang, B. Sun, F. Zhou, L. Sun, H. Zhao, X. Hu, X. Peng, J. Yan, H. Wang, W. Wang, J. Li, B. Yan, D. Kuang, Y. Wang, L. Wang, S. Duan, Volatile and nonvolatile memristive devices for neuromorphic computing, *Adv. Electron. Mater.* 8 (7) (2022) 2101127, <https://doi.org/10.1002/aelm.202101127>.
- [15] C. Yang, X. Wang, Full-circuit implementation of transformer network based on memristor, *IEEE Trans. Circuits Syst.* 69 (4) (2022) 1395–1407.
- [16] W. Yi, K.K. Tsang, S.K. Lam, X. Bai, J.A. Crowell, E.A. Flores, Biological plausibility and stochasticity in scalable vo2 active memristor neurons, *Nat. Commun.* 9 (1) (2018) 1–10, <https://doi.org/10.1038/s41467-018-07052-w>.
- [17] M. Ding, X. Wang, Z. Zeng, Memristive circuit design and implementation of olfactory perception and cross-modal association, *Sci. Sin. Inf.* 53 (12) (2023) 2561, <https://doi.org/10.1360/SSI-2023-0137>.
- [18] Y. Zhang, Z. Zeng, A multi-functional memristive Pavlov associative memory circuit based on neural mechanisms, *IEEE Trans. Biomed. Circuits Syst.* 15 (5) (2021) 978–993, <https://doi.org/10.1109/TBCAS.2021.3108354>.
- [19] Z. Chen, X. Wang, C. Yang, Z. Wang, Z. Zeng, Memristive circuit design for personalized emotion generation with memory and retrieval functions, *IEEE Trans. Cogn. Dev. Syst.* 16 (5) (2024) 1, <https://doi.org/10.1109/TCDS.2023.3317066>.
- [20] Z. Wang, X. Wang, A novel memristor-based circuit implementation of full-function Pavlov associative memory accorded with biological feature, *IEEE Trans. Circuits Syst. I Regul. Pap.* 65 (7) (2018) 2210–2220, <https://doi.org/10.1109/TCSI.2017.2780826>.
- [21] J. Sun, J. Han, Y. Wang, P. Liu, Memristor-based neural network circuit of emotion congruent memory with mental fatigue and emotion inhibition, *IEEE Trans. Biomed. Circuits Syst.* 15 (3) (2021) 606–616, <https://doi.org/10.1109/TBCAS.2021.3090786>.
- [22] M. Jiang, Z. Zeng, Memristive bionic memory circuit implementation and its application in multisensory mutual associative learning networks, *IEEE Trans. Biomed. Circuits Syst.* 18 (2) (2024) 308–321, <https://doi.org/10.1109/TBCAS.2023.3324574>.
- [23] X. Ji, S. Hao, K.G. Lim, S. Zhong, R. Zhao, Artificial working memory constructed by planar 2D channel memristors enabling brain-inspired hierarchical memory systems, *Adv. Intell. Syst.* 4 (3) (2022) 2100119, <https://doi.org/10.1002/aisy.202100119>.
- [24] J. Liu, F. Xiong, Y. Zhou, S. Duan, X. Hu, Bioinspired memristive neural network circuit design of cross-modal associative memory, *IEEE Trans. Cogn. Dev. Syst.* (2023) 1, <https://doi.org/10.1109/TCDS.2023.3303653>.
- [25] C. Yang, X. Wang, Z. Chen, S. Zhang, Z. Zeng, Memristive circuit implementation of operant cascaded with classical conditioning, *IEEE Trans. Biomed. Circuits Syst.* 16 (5) (2022) 926–938, <https://doi.org/10.1109/TBCAS.2022.3204742>.
- [26] J. Sun, L. Zhao, S. Wen, Y. Wang, Memristor-based neural network circuit of emotional habituation with contextual dependency, *IEEE Internet Things J.* 10 (19) (2023) 17382–17391, <https://doi.org/10.1109/IJOT.2023.3274116>.
- [27] C. Yang, X. Wang, Z. Chen, Z. Wang, S. Zhang, Z. Zeng, A bio-inspired decision-making memristive circuit based on classical and operant conditioning, *IEEE Trans. Autom. Sci. Eng.* (2023) 1–13, <https://doi.org/10.1109/TASE.2023.3326461>.
- [28] L. Shams, A.R. Seitz, Benefits of multisensory learning, *Trends Cogn. Sci.* 12 (11) (2008) 411–417, <https://doi.org/10.1016/j.tics.2008.07.006>.
- [29] C. Frings, C. Beste, E. Benini, M. Möller, D. Dignath, C.G. Giesen, B. Hommel, A. Kiesel, I. Koch, W. Kunde, et al, Consensus definitions of perception-action-integration in action control, *Commun. Psychol.* 2 (1) (2024) 7.
- [30] A. Baddeley, Working memory: theories, models, and controversies, *Annu. Rev. Psychol.* 63 (2012) 1–29, <https://doi.org/10.1146/annurev-psych-120710-100422>.
- [31] P. Domenech, E. Koehlin, Executive control and decision-making in the prefrontal cortex, *Curr. Opin. Behav. Sci.* 1 (2015) 101–106, <https://doi.org/10.1016/j.cobeha.2014.10.007>.
- [32] R. Lin, X. Meng, F. Chen, X. Li, O. Jensen, J. Theeuwes, B. Wang, Neural evidence for attentional capture by salient distractors, *Nat. Hum. Behav.* 8 (5) (2024) 932–944.
- [33] H. Atilgan, S.M. Town, K.C. Wood, G.P. Jones, R.K. Maddox, A.K. Lee, J.K. Bizley, Integration of visual information in auditory cortex promotes auditory scene analysis through multisensory binding, *Neuron* 97 (3) (2018) 640–655.e4, <https://doi.org/10.1016/j.neuron.2017.12.034>.
- [34] B.A. Anderson, The attention habit: how reward learning shapes attentional selection: the attention habit, *Ann. N. Y. Acad. Sci.* 1369 (1) (2016) 24–39, <https://doi.org/10.1111/nyas.12957>.
- [35] K. Dunovan, C. Vich, M. Clapp, T. Verstynen, J. Rubin, Reward-driven changes in striatal pathway competition shape evidence evaluation in decision-making, *PLoS Comput. Biol.* 15 (5) (2019) e1006998, <https://doi.org/10.1371/journal.pcbi.1006998>.
- [36] K.R. Ridderinkhof, W.P. van den Wildenberg, S.J. Segalowitz, C.S. Carter, Neurocognitive mechanisms of cognitive control: the role of prefrontal cortex in action selection, response inhibition, performance monitoring, and reward-based learning, *Brain Cogn.* 56 (2) (2004) 129–140, <https://doi.org/10.1016/j.bandc.2004.09.016>.
- [37] L. Scocchia, M. Val-Vecchi, J. Trietsch, Top-down influences on ambiguous perception: the role of stable and transient states of the observer, *Front. Hum. Neurosci.* 8 (2014) <https://doi.org/10.3389/fnhum.2014.00979>.
- [38] R. Yan, Q. Hong, C. Wang, J. Sun, Y. Li, Multilayer memristive neural network circuit based on online learning for license plate detection, *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* 41 (9) (2022) 3000–3011, <https://doi.org/10.1109/TCAD.2021.3121347>.
- [39] J. Li, Homeostatic synaptic plasticity as a metaplasticity mechanism — a molecular and cellular perspective, *Curr. Opin. Neurobiol.* 54 (2019) 44–53, <https://doi.org/10.1016/j.conb.2018.08.010>.
- [40] A. Baddeley, G. Hitch, R. Allen, Working memory and binding in sentence recall, *J. Mem. Lang.* 61 (3) (2009) 438–456, <https://doi.org/10.1016/j.jml.2009.05.004>.
- [41] H. Fang, Y. Hu, L. Yang, Y. Liu, The role of phonological loop and visuospatial sketchpad in virtual maze wayfinding, *J. Environ. Psychol.* 67 (2020) 101378, <https://doi.org/10.1016/j.jenvp.2019.101378>.
- [42] Y. Zhang, Z. Zeng, Brain-inspired model and neuromorphic circuit implementation for feature-affective associative memory network, *IEEE Trans. Cogn. Dev. Syst.* 16 (5) (2024) 1–15, <https://doi.org/10.1109/TCDS.2023.3329044>.
- [43] S. Funahashi, J.M. Andreau, Prefrontal cortex and neural mechanisms of executive function, *J. Physiol.-Paris* 107 (6) (2013) 471–482, <https://doi.org/10.1016/j.jphysparis.2013.05.001>.
- [44] F. Collette, M. Van der Linden, Brain imaging of the central executive component of working memory, *Neurosci. Biobehav. Rev.* 26 (2) (2002) 105–125.
- [45] S. Balatti, S. Ambrogio, Z. Wang, D. Ielmini, True random number generation by variability of resistive switching in oxide-based devices, *IEEE J. Emerg. Sel. Top. Circuits Syst.* 5 (2) (2015) 214–221, <https://doi.org/10.1109/JETCAS.2015.2426492>.

Author biography



Jihong Zhang received her B.E. degree from Chongqing University in 2023. She is currently working toward the M.E. degree at the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, China. Her current research interests include brain-inspired bionic memristive circuit design.



Xiaoping Wang (SM'19) received her B.S. and M.S. degrees in automation from Chongqing University, Chongqing, China, in 1997 and 2000, respectively, and the Ph.D. degree in systems engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2003. Since 2011, she has been a Professor at the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology. Her current research interests include memristors and their application to memory storage, modeling, and simulation.



Zhanfei Chen received his B.E. degree from Huazhong University of Science and Technology in 2020. He is currently working toward the Ph.D. degree at the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, China. His current research interests include brain-inspired bionic memristive circuit design.



Chao Yang received his B.S. degree in electronic information engineering and his M.S. degree in electronic science and technology from the School of Electronic Engineering of Guangxi Normal University, Guilin, China, in 2012 and 2015 respectively, and the Ph.D. degree in artificial intelligence at the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, China, in 2024. He is currently a lecturer with the School of Computer Science and Technology, Guangxi University of Science and Technology, Liuzhou, China. His current research interests include memristive systems and circuits, memristive neural networks, and brain-like intelligent computing circuits based on memristor.